

SFTP User Manual

SFTP Enterprise Management System — User Manual

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This guide is for SFTP end users – people who connect to upload, download, and manage files on the enterprise SFTP server. If you are a system administrator looking for account management, see the [Administrator Guide](#) instead.

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1. What is this SFTP service?

The SFTP Enterprise Management System provides you with a secure, private space on a remote server where you can upload, download, and manage files. SFTP (SSH File Transfer Protocol) encrypts all data in transit – your files, your password, and every command you send are protected from eavesdropping.

Your administrator creates an account for you with a username, password, and permission level. Once your account is active, you connect from any standard SFTP client and work with your files inside your own home directory. The service is powered by the well-established atmoz/sftp container, backed by encryption at rest for all login credentials.

2. How to get access

Access is granted by your system administrator. The process is simple:

1. **Ask your admin** to create an SFTP account for you. They will need to choose:
 - A **username** for you (lowercase, letters/digits/underscores/dots/hyphens only).
 - A **password** (12 characters minimum, strong).
 - A **permission level** (read-write for full access, or read-only if you only need to download).
 2. Your admin will provide you with:
 - The **server address** (hostname or IP).
 - The **port number** (default: 7721).
 - Your **username** and **password**.
 3. Your home directory is automatically created under your username (e.g., /alice/). The only writable location is the upload/ subdirectory inside your home – this is by design for security.
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3. Connecting via SFTP client

You can use any standard SFTP client. Here are the three most common options.

Command-line sftp (Linux / macOS)

The sftp command is built into every Linux and macOS system – no installation needed.

```
# Connect (replace with your actual username and server address)
sftp -P 7721 your_username@server_address
```

```
# Example: alice connecting to sftp.example.com
sftp -P 7721 alice@sftp.example.com
```

When prompted, enter your password. Once connected you will see the `sftp>` prompt. Type `help` for a list of available commands, `exit` to disconnect.

FileZilla (Windows / macOS / Linux)

FileZilla is a free, graphical SFTP client available at <https://filezilla-project.org/>.

Setup steps:

1. Download and install FileZilla from the official website.
2. Open FileZilla. Click **File > Site Manager**.
3. Click **New Site** and configure:
 - **Protocol:** SFTP - SSH File Transfer Protocol
 - **Host:** Your server address (e.g., `sftp.example.com`)
 - **Port:** 7721
 - **Logon Type:** Normal
 - **User:** Your username
 - **Password:** Your password
4. Click **Connect**.

Once connected, your local files appear on the left panel and remote files on the right. Drag and drop files between them.

WinSCP (Windows)

WinSCP is another free, popular SFTP client for Windows at <https://winscp.net/>.

Setup steps:

1. Download and install WinSCP.
2. On the login screen:
 - **File protocol:** SFTP
 - **Host name:** Your server address
 - **Port number:** 7721
 - **User name:** Your username
 - **Password:** Your password
3. Click **Login**.

Using SSH keys instead of passwords

If your administrator has configured your account with an SSH public key, you can connect without typing a password:

```
# Command-line: use the -i flag to point to your private key
sftp -P 7721 -i ~/.ssh/id_ed25519 alice@sftp.example.com

# FileZilla: in Site Manager, set Logon Type to "Key file" and
              browse to your private key file.
# WinSCP: click "Advanced" > "SSH" > "Authentication" and browse
           to your private key file.
```

SSH keys are more secure than passwords and recommended for automated scripts and frequent access.

4. Uploading files

Files are uploaded to the upload/ subdirectory inside your home. The top level of your home directory is read-only (by design, for security).

Command-line sftp:

```
sftp -P 7721 alice@sftp.example.com
# Once connected:
sftp> cd upload
sftp> put /path/to/your/local/file.txt
# To upload an entire directory:
sftp> put -r /path/to/your/local/folder/
```

FileZilla: Browse to the upload/ folder in the right panel (remote site), then drag files from the left panel (local site) into it. You can also right-click a local file and select **Upload**.

WinSCP: Navigate to the upload/ folder on the remote side, then drag files from your local panel, or press **F5** to copy selected files to the remote side.

Important notes:

- You can only upload to the upload/ subdirectory. The top level of your home is intentionally read-only.
 - Uploaded files are visible to other users with access to your home directory only if your admin set up a shared/public arrangement.
 - Large files may take time – SFTP shows a progress bar in graphical clients.
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5. Downloading files

Downloading works the same way in reverse. You can download any file you can see.

Command-line sftp:

```
sftp -P 7721 alice@sftp.example.com
# Once connected:
sftp> cd upload
sftp> get remote_file.txt
# To download to a specific local path:
sftp> get remote_file.txt /home/alice/Downloads/
# To download an entire directory:
sftp> get -r some_folder/
```

FileZilla: Browse to the file in the right panel (remote site), right-click it and select **Download**, or drag it to the left panel.

WinSCP: Select the file on the remote side and press **F5**, or drag it to your local panel.

6. Permission levels explained

Your administrator assigns one of two permission levels to your account. This determines what you can do.

Read-only (read_only)

With read_only permission you can:

- **List** files and directories (see what is there).
- **Download** files (copy them to your local machine).

You **cannot**:

- Upload new files.
- Delete or rename existing files.
- Create new directories.

Attempting any write operation will result in a “**Permission denied**” error. This is expected behavior when your account is configured for download-only access.

Read-write (read_write)

With read_write permission you can:

- **List** files and directories.

- **Download** files.
- **Upload** new files (to the upload/ subdirectory).
- **Delete** files (inside your writable subtree).
- **Rename** files.
- **Create** subdirectories.

This is the full-access permission. Your admin will assign this if you need to both send and receive files.

7. Public shares

Your administrator may create a **public** account – one that anyone can access without a password (or with a shared, well-known credential). Public accounts are always **read-only**: they can download files but never upload, delete, or modify anything.

Public access is **never enabled by default** – your administrator must explicitly acknowledge and confirm this setting. If there is a public share available to you, your admin will provide the connection details.

Typical use cases for public shares:

- A company-wide download portal for shared resources, documentation, or software installers.
 - A read-only drop for contractors or partners who only need to receive files.
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8. Understanding your directory layout

When your account is created, your home directory follows this structure:

```
data/  
  your_username/          # Your home (chroot target -- you cannot  
navigate above this)  
  upload/                 # The writable area (for read_write  
users)
```

Key points:

- You are **confined to your home directory**. You cannot browse to other users' directories or system files.
- The **top level** of your home (/your_username/) is read-only. You cannot create files here. This is an OpenSSH security requirement – the chroot target must not be writable by the user.

- The **upload/ subdirectory** is where you place and retrieve files. This is the working directory you should use.
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9. Troubleshooting common issues

“Connection refused”

What it means: The SFTP server is not reachable at the address and port you specified.

Check: - Is the server address correct? Ask your admin for the exact hostname or IP. - Is the port correct? The default SFTP port is 7721 - not the standard SSH port 22. - Is the server running? The server may be down for maintenance. Contact your admin.

“Permission denied” when uploading

What it means: Your account does not have write permission, or you are trying to upload to a read-only location.

Check: - Are you in the upload/ subdirectory? You cannot upload to the top level of your home. - Is your account `read_only`? If so, you cannot upload at all. Ask your admin to upgrade your permission to `read_write` if you need write access. - Are you uploading to a public share? Public accounts are always read-only.

“Permission denied” when connecting

What it means: Your credentials (username or password) are incorrect, or your account is disabled.

Check: - Verify your username and password with your admin. Passwords are case-sensitive. - Ask your admin whether your account is enabled. Disabled accounts cannot log in. - If using SSH keys: verify the key file path is correct and that your public key was correctly registered by your admin.

“No such file or directory”

What it means: The remote path you specified does not exist.

Check: - Did you `cd upload` before trying to access your files? Remember, your working directory on first connect is your home. - Use `ls` or `dir` to list what files and directories exist.

“Connection timed out”

What it means: The server did not respond within the expected time.

Check: - Is a firewall blocking port 7721? You may need to be on a VPN or specific network. Contact your admin. - Is the server address reachable? Try `ping <server_address>` to verify basic network connectivity.

Files appear empty or corrupted after upload

What it means: The upload may have been interrupted.

Check: - Re-upload the file. SFTP does not automatically resume interrupted transfers. - For large files, ensure your connection is stable throughout the upload. - Verify the file size matches after upload: compare local `ls -l` output with remote `ls -l`.

10. Screenshots (to be added)

Screenshots of the connection and file-transfer process will be added in a subsequent release. This section will include:

- FileZilla Site Manager configuration screen showing the SFTP settings.
 - FileZilla main interface with local and remote file panels.
 - WinSCP login screen with the SFTP protocol selected.
 - Command-line sftp session showing a successful login, upload, and download.
 - Common error messages and their resolutions in graphical clients.
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Sources verified (2026-07-11)

- Internal: docs/guides/quick_setup_guide.md – port configuration (7721 for SFTP, 7722 for API), account creation workflow, permission levels, end-to-end verification procedure.
- Internal: qa/results/MANUAL-QA-report.md – confirmed all API endpoints work correctly against the live `sftp-0.1.0-dev-0.1.0` binary: health, login, refresh, logout, account CRUD, sync, public-guard.
- Internal: docs/architecture/permissions_model.md – confirmed permission enum (`read_only`, `read_write`), public-never-default guard, chroot directory layout, enforcement points.

- External: atmoz/sftp Docker Hub (<https://hub.docker.com/r/atmoz/sftp>) - users.conf grammar, :e encrypted password marker, chroot home-not-writable requirement.
- External: FileZilla (<https://filezilla-project.org/>) and WinSCP (<https://winscp.net/>) official documentation for client setup instructions.